

a) Označme $B = (f, g, h)$ ortonormálnu bázu podpriestoru $\text{span}\{1, x, x^2\}$.

Podľa Gram-Schmidtovj ortogonalizácie:

$$f = \frac{1}{\|1\|} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 dx}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$g' = x - \text{pr}_{\text{span}\{f\}} x = x - \langle f, x \rangle f = x - \left(\int_1^5 \frac{1}{2} x dx \right) \frac{1}{2} = x - 6 \cdot \frac{1}{2} = x - 3$$

$$g = \frac{g'}{\|g'\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$h' = x^2 - \text{pr}_{\text{span}\{f, g\}} x^2 = x^2 - \langle f, x^2 \rangle f - \langle g, x^2 \rangle g =$$

$$x^2 - \left(\int_1^5 \frac{1}{2} x^2 dx \right) \frac{1}{2} - \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 dx \right) \left(\frac{\sqrt{3}}{4} (x-3) \right) =$$

$$= x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \left(\frac{\sqrt{3}}{4} (x-3) \right) = x^2 - 6x + \frac{23}{3}$$

$$h = \frac{h'}{\|h'\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{\sqrt{45}}{16} \left(x^2 - 6x + \frac{23}{3} \right)$$

$$\text{Teda } B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{\sqrt{45}}{16} (x^2 - 6x + \frac{23}{3}) \right)$$

$$b) \text{pr}_{\text{span}\{f, g, h\}} \sin x = \langle f, \sin x \rangle f + \langle g, \sin x \rangle g + \langle h, \sin x \rangle h =$$

$$= \left(\int_1^5 \frac{\sin x}{2} dx \right) \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx \right) \frac{\sqrt{3}}{4} (x-3) + \left(\int_1^5 \frac{\sqrt{45}}{16} (x^2 - 6x + \frac{23}{3}) \sin x dx \right) \frac{\sqrt{45}}{16} (x^2 - 6x + \frac{23}{3}) =$$

$$\begin{aligned} \text{(zobrohlenie na} \\ \text{2 desatinne miesta)} \end{aligned} \quad \doteq 0,73 \cdot \frac{1}{2} - 1,49 \cdot \frac{\sqrt{3}}{4} (x-3) - 1,78 \cdot \frac{\sqrt{45}}{16} (x^2 - 6x + \frac{23}{3}) \doteq$$

$$\doteq -0,75x^2 + 3,83x - 3,72$$

$$c) \|\sin x - (-0,75x^2 + 3,83x - 3,72)\| = \sqrt{\int_1^5 (\sin x + 0,75x^2 - 3,83x + 3,72)^2} \doteq 1,70$$